Speech Technology

Section Id :	64065345320
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	4
Number of Questions to be attempted :	4
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065396859
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 123 Question Id : 640653676903 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS

REGISTERED BY YOU)

Options :

6406532266816. ✔ YES

6406532266817. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065396860
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 124 Question Id : 640653676904 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

1. (1 point) In K-means, the number of clusters (K) changes as the algorithm reaches to convergence.

   A. True

   B. False

No, this happens in LBG.

2. (1 point) A uni-variate Gaussian distribution is completely defined by the following parameters.

   A. mean, variance

   B. median, covariance

   C. mode, variance

   D. mode, covariance

3. (1 point) K-Means will always find the global minimum.

   A. True

   B. False

it is an iterative greedy optimizan that depends on the initializan. It might get stuck somewhere in local minima.

4. (3 points) Consider the set of training data below, and two clustering algorithms: K-Means and a Gaussian Mixture Model (GMM) trained using EM. Will these two clustering algorithms produce the same cluster centers (means) for this data set? Explain why or why not.

Options :

6406532266818. ✔ I have written answers on the answer sheets

6406532266819. ✖ Not applicable

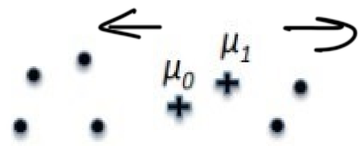
Sub-Section Number :	3
Sub-Section Id :	64065396861
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 125 Question Id : 640653676905 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 11

Question Label : Multiple Choice Question

5. (3 points) Assume there are two Gaussian components in the GMM; μ_0, μ_1, σ_0 and σ_1 define means and variances of these two components, π_0 and $(1-\pi_0)$ denote the mixture proportions of the two Gaussians (i.e. $p(x) = \pi_0 N(\mu_1, \sigma_1) + (1-\pi_0) N(\mu_2, \sigma_2)$).



(a) (1 point) Draw on the figure the directions in which µ0 and µ1 will move during the next M-step.

(b) (2 points) Will the estimate of π_0 increase or decrease on the next EM step? Explain your reasoning in one sentence. π_0 is the expected proporⁿ of pts. assigned to comp. 0. $\pi_0 \downarrow$ as it will be responsible for fewer of the total pts.

6. (5 points) We saw two approaches to solve the sequence prediction problem: greedy and viterbi. Consider a weather sequence prediction problem, assuming first-order markov chain. Given that Sunny weather was observed on Day-0, compute the best possible weather forcast sequence for the next three days using 1) greedy approach, 2) viterbi approach.

Hint: For viterbi keep a track of path taken to calculate $\gamma_A(B)$, where $\gamma_A(B)$ denotes maximum probability of reaching state A at time step B.

$\overset{0}{S} \rightarrow \overset{1}{S} \rightarrow \overset{2}{S} \rightarrow \overset{3}{S} \Rightarrow$ greedy

S

$\gamma_S(1)$ 0.5 $\xrightarrow{0.5}$ $\gamma_S(2)$ 0.25 $\xrightarrow{0.5}$ $\gamma_S(3)$ 0.125

$\gamma_C(1)$ 0.375, $\gamma_C(2)$ 0.19, $\gamma_C(3)$ 0.0975

$\gamma_R(1)$ 0.125, $\gamma_R(2)$ 0.23, $\gamma_R(3)$ 0.155 25

0.375, 0.625

Today

		Sunny	Cloudy	Rainy
Yesterday	Sunny	0.5	0.375	0.125
	Cloudy	0.25	0.125	0.625
	Rainy	0.25	0.075	0.675

7. (3 points) We saw two approaches for vector quantization, namely: k-means & Linde-Buzo-Gray Algorithm (LBG). Explain in detail the clustering mechanism used in these two algorithms, highlighting differences in their approach.

k means: fixed no. of clusters | dynamic no. of

Options :

6406532266820. ✔ I have written answers on the answer sheets

6406532266821. ✖ Not applicable

Sub-Section Number :	4
Sub-Section Id :	64065396862
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 126 Question Id : 640653676906 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8

Question Label : Multiple Choice Question

8. (1 point) For perfect reconstruction of a bandlimited signal, the sampling frequency must be _____.

   A. equal to twice the maximum frequency

   B. greater than equal to twice the maximum frequency

   C. lesser than equal to twice the maximum frequency

   D. none of the above

9. (3 points) For each of the following discrete time system, comment upon the following system properties (1) linear/non-linear, (2) time-invariant/time-variant, and (3) causal/anti-causal.

   (a) (1.5 points) $y[n] = nx[n]$ → linear, TV,

   (b) (1.5 points) $y[n] = e^{x[n]}$

   → non-linear, time I & causal

10. (1 point) Any point in a 3-D space can be represented as a _____ combination of its basis functions. linear

11. (1 point) Let $f_X(x)$ be the probability distribution function for random variable X then

$$\int_{-\infty}^{\infty} f_X(x)dx = ?$$ 1

   A. 1

   B. ∞

   C. 0

   D. none of the above

12. (1 point) The Gaussian curve is always symmetrical about _____.

   A. 1 — mean

   B. 0

   C. standard deviation

   D. mean

13. (1 point) Psychophysical equivalent of frequency is pitch.

Options :

6406532266822.

✔ I have written answers on the answer sheets

6406532266823. ✖ Not applicable

Programming in C

Section Id :	64065345321
Section Number :	9
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	19
Number of Questions to be attempted :	19
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065396863
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 127 Question Id : 640653676907 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : PROGRAMMING IN C (COMPUTER